EVIDENCE FOR HEAT-TREATED STONE TOOLS: MAGNETIC MINERALOGY OF HEATED CHERT

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1. MOTIVATION

Intentional heating of geological materials for tool-making purposes has been observed at Stone Age archaeological sites in many parts of the world, including many regions of North America. Studies on detecting heat-treated materials have been a popular subject among archaeologists, and a variety of specialized and expensive approaches have been developed. Magnetic methods have not yet been adapted to this problem, yet they offer numerous potential advantages over existing techniques in that they are fast, non-destructive, and inexpensive. The goal of this study is to compare the magnetic properties of unheated and heated chert, so that we may determine if there are diagnostic signals that indicate whether a chert has been intentionally heated.

2. METHODOLOGY

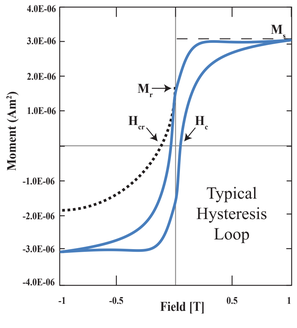
Eleven different chert cobbles were collected from various localities throughout the Midwest that are known to be common sources for Stone Age tool manufacture.

Low Field Susceptibility

Magnetic Susceptibility (χ) is a sample's magnetization within a low induced magnetic field, and is usually interpretted as a rough measure of the amount of magnetic material in a sample. Chert primarily consists of the mineral quartz, which is "diamagnetic" and produces low negative c values. Trace amounts of clay minerals in chert are "paramagnetic" and produce small positive χ values. Trace amounts of iron oxides, like magnetite and hematite, are "ferromagnetic" and produce dramatically larger positive χ values.

Remanence

Magnetic "remanence" is the magnetization permanently recorded by a sample after an external field has been removed.



Hysteresis

Saturation Magnetization (**Ms**) : The maximum magnetization produced in the presence of an applied field.

Saturation Remanent Magnetization (**Mr**): The maximum possible magnetic recording maintained by a sample after exposure to a saturating magnetic field.

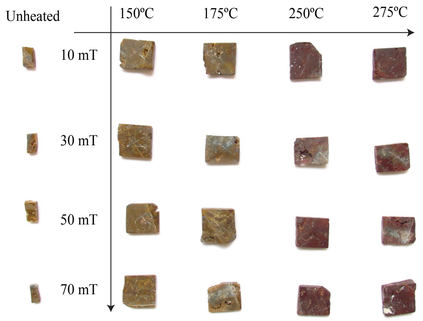
Coercivity (**Hc**): The field required to reverse half of a sample's induced magnetization.

Coercivity of Remanence (**Hcr**): The magnetic field required to flip half of a sample's remanence such that its net moment is zero.

The ratios of Mr/Ms and Hcr/Hc are useful tools for determining the grain size of magnetic minerals

3. RESULTS

3.1 Change in Color



	Unheated	Heated
Burlington Chert	Light grey	Pale white
Cochrane Chert	Brown with grey inclusions	Dark red brown
Cedar Valley Chert	Light yellow brown	Dark red brown
Galena Chert	Light grey	Light Pink
Jasper Taconite	Black with red inclusions	No Change
Knife Lake Siltstone	Black	No Change
Grand Meadow Chert	Black	No Change
Prairie du Chien Chert	Light grey	Pale white
Knife River Flint	Yellow brown to dark grey	No Change
Tongue River Silica Grey	Grey with red inclusions	Light grey
Tongue River Silica Red	Red brown with grey inclusion	Red

Figure 1. Color change for the Cochrane Chert. Each tile was heated at a different temperature and field combination, but similar temperature treatments produced repeatable color changes.

7 out of the 11 chert sources exhibited a color change due to heating. Darker raw samples showed little to no color change. Some localities, such as the Galena Chert, required a higher temperature to produce a color change (>325°C); whereas, other samples changed colors at relatively lower temperatures (~200°C).

3.2 Change in Magnetic Mineral Concetration

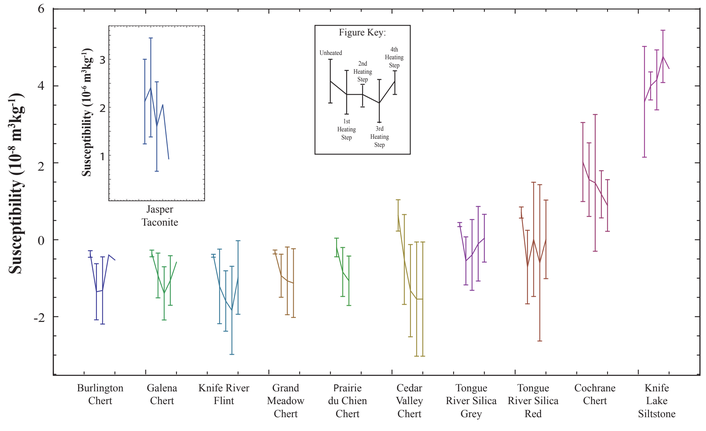


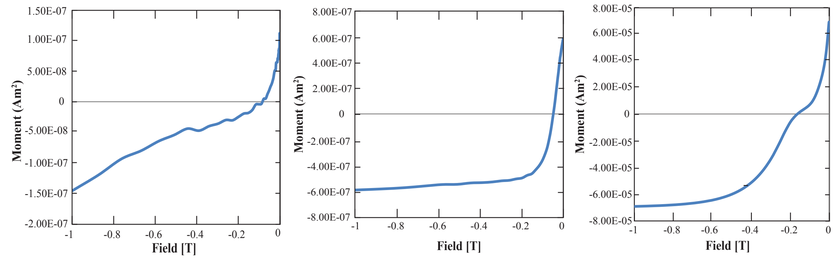
Figure 2. Change in susceptibility as a function of heating for 11 different chert source locations. Each locality contained 20 samples with 4 different heat treatments. Error bars represent one standard deviation. Points without error bars contain only one sample. Additional data is still being collected.

A general decreasing trend in susceptibility after heating is seen in all but one location. In those localities, susceptibility levels off or suddenly increases at higher temperatures when new magnetic minerals are produced. The 2nd most magnetic sample, Knife Lake Siltstone, is the only sample that shows a systematic increase in susceptibility upon heating.

3.3 Changes in Magnetic Mineral Composition

Variability in Composition

"Back-field curves" measure the remanence acquired by a sample after exposure to progressively larger negative fields. The shapes of these curves tell us about the magnetic mineralogy in a sample:



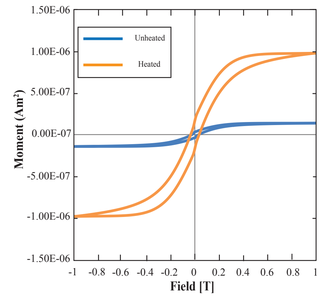
Sample that contains a large amount of hematite

Sample that contains a large amount of magnetite

Sample that contains a mixture of hematite and magnetite

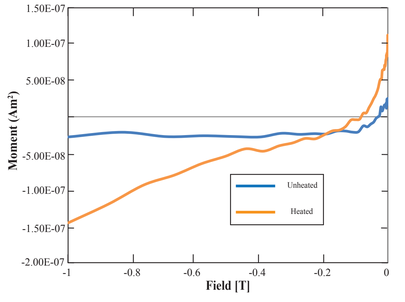
Before and After Heating

Hysteresis Loop



Hysteresis loops show an increase in Ms, Mr, and Hc upon heating, suggesting that a new high coercivity mineral was created.

Back-field Curve



Back-field curves help identify which new minerals were created in the heated samples. In most cases, we see a increase in the relative abundance of hematite.

4. CONCLUSIONS

In this study, we observed that heating chert changes the physical appearance and magnetic characteristics of the stone. In most sources, there was a reddening of the samples and an increase of luster. Most samples showed a decrease in susceptibility after heating, except in one locality. The susceptibility data and hysteresis plots show that magnetic mineral compositions and concentrations were changing considerably after heating. In most cases, we hypothesize that the mineral hematite (Fe_2O_3) appears to precipitate during heating, while grains of magnetite (Fe_3O_4) become progressively more oxidized. Thus, by comparing the magnetic properties of a chert artifact to those of an unheated sample from the same geologic source, one should be able to determine whether or not the artifact had undergone heat treatment.